CSCI 3141
Foundations of Data Science in R *(and Python)*

DALHOUSIE UNIVERSITY

Course Orientation & Data Science Process

What data science is and how a project runs

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1 Why learn data science

What it is for This section explains why data skills are in demand and what has made the field grow.

1.1 The labour market

Indeed's 2022 ranking of the twenty best jobs in Canada, across all fields, is shown in Table 1. Two of the twenty are data roles. Indeed ranked roles by employer demand, a base salary of at least $75,000, and growth in each role's share of job postings. The table is a historical example of how posting growth and salary describe different aspects of a job market. It is not a current employment forecast.

Table 1. Historical 2022 ranking of top jobs in Canada (Indeed), with the two data roles in bold

Rank	Job	Average Salary	Share Growth (%), 2019–22
1	**Data engineer**	**$85,706**	**242**
2	Site reliability engineer	$90,010	218
3	Engineering manager	$111,515	161
…	ranks 4 to 18, fifteen further roles	…	
19	**Data scientist**	**$76,938**	**60**

Source: Indeed's 2022 ranking as reported by Canadian HR Reporter. Data engineering, first on this list, is the work of making data available and reliable. Much of what this course calls data preparation is the analyst's share of that work.

1.2 What is driving the field

Data volumes keep increasing. Cheap and varied sensors, wider networking, and the Internet of Things all generate records that are stored by default rather than discarded.

Computing power keeps increasing. Methods that were once impractical on a single machine now run on a laptop, which changes what is worth attempting.

Data science is also *interdisciplinary*, and it is applied in many domains, including healthcare, finance and banking, transportation and smart cities, business and e-commerce, social media and social network analysis, and sustainability and climate. Each domain supplies the questions and the domain knowledge needed to interpret an analysis.

2 Data and tabular data

What it is for Every method in this course starts from a table. This section fixes the words used to describe one.

2.1 Types of data

Data volumes are growing for several types of data.

- Tabular data.
- Databases, that is, multiple related data tables.
- Text.

- Multimedia, such as images, video, and sound.
- Data streams and time series.

This course works with **tabular data**. Almost every idea in the course, from distance to model assessment, is defined first on a table. The definitions carry over to other data types once those are given a tabular representation.

2.2 The structure of a table

Definition

A data set is a collection of observations, written as an $n \times p$ *data matrix* with n observations or cases, each described by p variables. Entry x_{ij} is the value of variable j for case i.

Figure 1 shows where x_{ij} sits in the matrix.

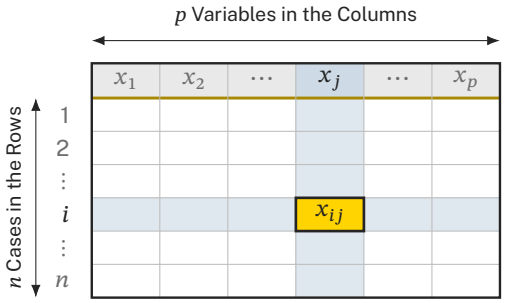


Figure 1. An $n \times p$ data matrix. Row i is one case and column j is one variable, and x_{ij} is the cell where they cross.

- Rows represent the cases. They are also called objects, examples, instances, or records.
- Columns represent variables. They are also called features, fields, or attributes.
- A *numeric* variable measures a quantity. A *categorical* variable records a label.
- An *identifier* names a record. It is not a measurement and should not be averaged.

The *observational unit* is what one row describes, for example one patient, one household, or one sensor reading. State it before counting observations.

2.3 The example data set

The running example is the Heart Disease data set, with **12 attributes** and **918 cases**. Its Kaggle page is titled *Heart Failure Prediction Dataset*. Cite it as fedesoriano (2021), *Heart Failure Prediction Dataset*, Kaggle [4]. It combines five heart-disease databases from the UCI Machine Learning Repository [6]. The first six rows are shown in Table 2.

Here $n = 918$ and $p = 12$. One row is one patient, so the patient is the observational unit. Seven variables are categorical and five are numeric, namely the four integer-valued measurements `Age`, `RestingBP`, `Cholesterol` and `MaxHR`, and `Oldpeak`. These are analytical roles, not the storage types returned by an import function. `FastingBS` and `HeartDisease` store categories as the integers 0 and 1. Numeric storage does not make a category continuous.

`HeartDisease` is the outcome that the predictive example will try to estimate. In the full table, 508 of the 918 patients have it, or 55%. The same 55% appears at the root of the decision tree in section 4.

Table 2. First six rows of the Heart Disease data set

Row	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS
	numeric	category	category	numeric	numeric	category
1	40	M	ATA	140	289	0
2	49	F	NAP	160	180	0
3	37	M	ATA	130	283	0
4	48	F	ASY	138	214	0
5	54	M	NAP	150	195	0
6	39	M	NAP	120	339	0

Row	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
	category	numeric	category	numeric	category	category
1	Normal	172	N	0.0	Up	0
2	Normal	156	N	1.0	Flat	1
3	ST	98	N	0.0	Up	0
4	Normal	108	Y	1.5	Flat	1
5	Normal	122	N	0.0	Up	0
6	Normal	170	N	0.0	Up	0

2.4 Importing the table

Download and extract `heart.csv` from the Kaggle dataset page. Place the CSV in your working directory, the folder from which your R or Python session reads files. The code below reads the table and inspects it.

R

```r
heart <- read.csv(
  "heart.csv",
  stringsAsFactors = TRUE
)
dim(heart)       # 918  12
head(heart)
str(heart)
table(heart$HeartDisease)
```

Python

```python
import pandas as pd

heart = pd.read_csv("heart.csv")
print(heart.shape)  # (918, 12)
print(heart.head())
print(heart.dtypes)
heart["HeartDisease"].value_counts()
```

Both imports produce 918 rows and 12 columns. `dim` and `shape` report the dimensions n and p. The `head` calls show the first rows, and `str` and `dtypes` show storage types. The final calls count the two outcome categories, 410 patients coded 0 and 508 coded 1.

Differences to know

- R assigns a value to a name with `<-`. Python uses `=`.
- R prints each top-level call. A notebook cell displays only the value of its last line, so the Python listing wraps earlier lines in `print`.
- R selects a column with `$`. Python uses bracket notation, `heart["HeartDisease"]`. Python reads tables through `pandas`, a library for tables.
- With `stringsAsFactors = TRUE`, R imports text as factors, its representation of categories. `pandas` 3 stores text with the `str` dtype (`object` in pandas 2). In both languages the integer-coded categories `FastingBS` and `HeartDisease` arrive as integers, and only the dataset documentation says they are labels.

3 What data science is

What it is for A working definition says what counts as data science and which skills it needs.

Definition

"Data mining is the analysis of (often large) observational data sets to find unsuspected relationships and to summarize the data in novel ways that are both understandable and useful to the data owner."

Hand, Mannila, and Smyth, *Principles of Data Mining*, MIT Press, 2001, page 1 [5].

The definition stresses discovering relationships and producing useful summaries. Data science combines that work with computing, statistical reasoning, and knowledge of the subject being studied. Donoho traces how the field grew out of statistics and why it reaches beyond it [3].

Observational means the researcher does not assign an experimental intervention. Such data may still be collected for a research question. An observed relationship alone does not establish cause and effect. Data science also uses experiments. The Heart Disease table is observational, because no patient was assigned a heart rate or an age.

Figure 2 places data science at the intersection of three disciplines, computer science and information technology, mathematics and statistics, and domain or business knowledge. The framing is widely attributed to Drew Conway (2010) [2], and the labels are this course's. Each pairwise intersection is a recognizable field. Machine learning sits between computing and statistics, software development between computing and the domain, and traditional research between statistics and the domain.

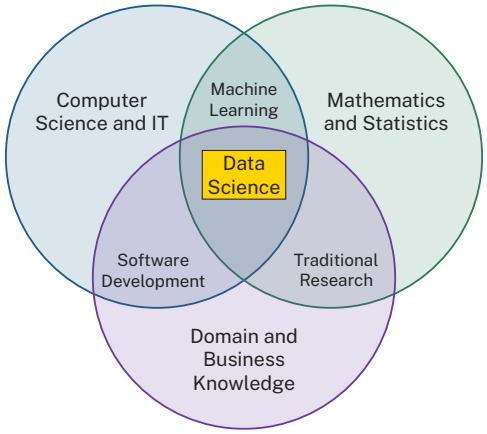


Figure 2. Data science at the intersection of computing, statistics, and domain knowledge. Three-circle framing after Drew Conway (2010) [2].

4 Data science tasks

What it is for Naming the kind of question fixes which methods and which results apply.

The three task families Figure 3 names each family, its sub-tasks, and the question it answers. A *target* is the variable a method is asked to predict. Predictive analytics uses one. Exploratory and descriptive analytics do not.

The three examples below answer each question on the Heart Disease data. Read the output of each one. Lab 01 uses descriptive statistics and plots only.

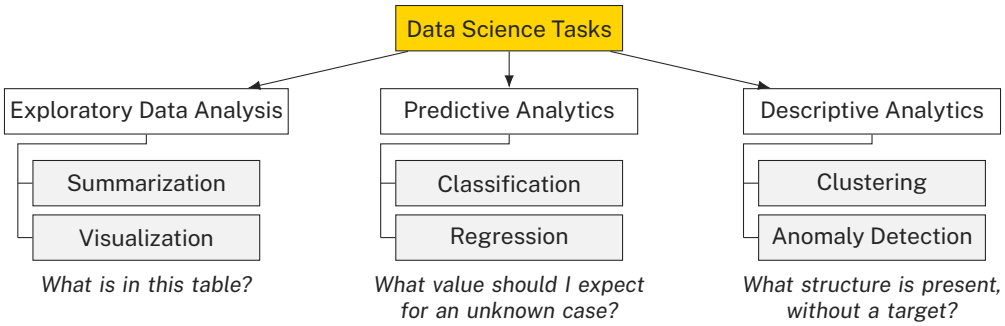


Figure 3. Three data science task families, their sub-tasks, and the questions they answer

4.1 Exploratory data analysis

Exploratory analysis summarizes and plots the table without fitting anything.

- A *histogram* divides a numeric variable into intervals, called bins, and draws one bar per bin whose height counts the observations in it.
- A *scatter plot* places one point at each pair of measurements.
- The *median* is the middle value after sorting, or the mean of the two middle values when the number of observations is even.

In Figure 4 the left panel is a histogram of `Age`, stacked by `HeartDisease`, with the median age of 54 marked. The right panel is a scatter plot of `MaxHR` against `Age`, coloured by `HeartDisease`. Overlapping points may hide patients. Together the panels show that patients with heart disease in this sample tend to be older and to reach a lower maximum heart rate. The figures describe the sample, and nothing has been fitted.

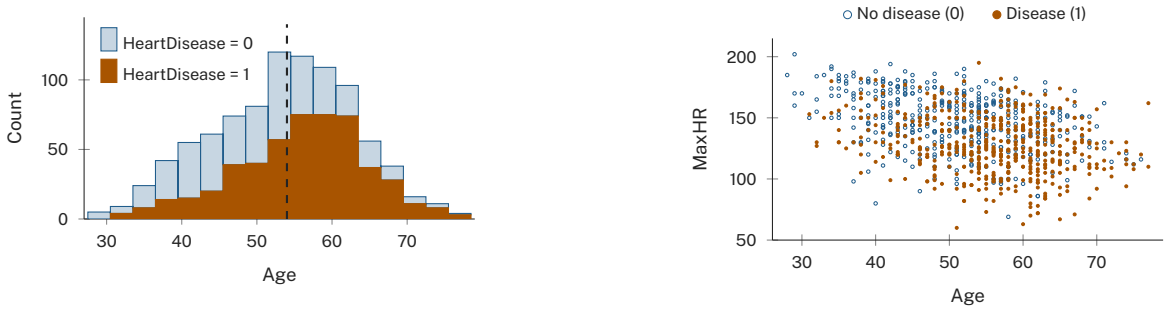


Figure 4. Exploratory views of all 918 patients. Three-year age bins start at 27.5; the dashed line marks the median age of 54. The scatter uses unjittered observations, so coincident points overlap. Neither plot fits a model.

4.2 Predictive analytics

Definitions

A *model* is a rule fitted to data. *Classification* predicts a category, and *regression* predicts a number. A *classification tree* expresses its rules as branches. The *training data* are the cases used to fit the rules.

Figure 5 shows a tree with `HeartDisease` as the target, fitted on `Age` and `MaxHR` only and limited to three leaves. Each node carries three values, the predicted class, the proportion of disease cases in the node, and the share of the training data that reaches the node.

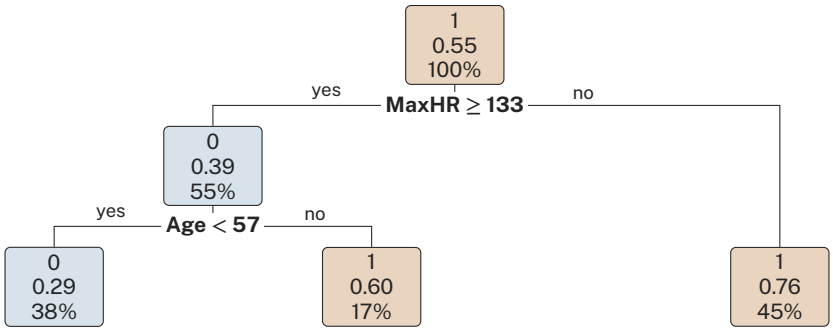


Figure 5. A depth-two classification tree for `HeartDisease`. Each node gives the predicted class, the proportion of disease cases, and the share of the training data reaching it. Rust nodes predict disease and blue nodes predict no disease.

Read the tree as a set of rules.

1. If `MaxHR` is below 133, predict disease. 76% of the patients in that leaf have it.
2. Otherwise, if `Age` is below 57, predict no disease.
3. Otherwise, predict disease.

The fitted boundaries are 132.5 and 56.5, which are equivalent to these integer rules on the recorded measurements. The two splits cut the `Age`–`MaxHR` plane into three rectangles, shown in Figure 6. The low-heart-rate region predicts disease, and above the heart-rate threshold age decides.

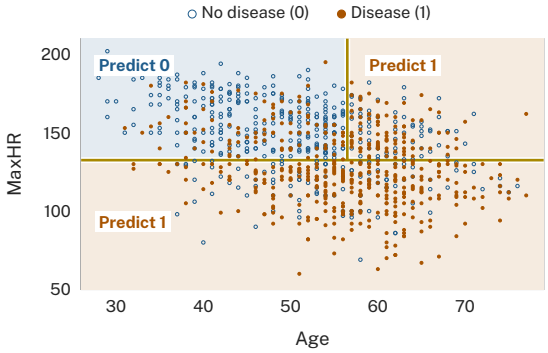


Figure 6. The same fitted tree partitions the `Age`–`MaxHR` plane. Blue regions predict no disease and rust regions predict disease; the points show the observed classes. The gold boundaries represent the two fitted splits.

Training accuracy Accuracy is the share of cases predicted correctly. The tree predicts 654 of the 918 training patients correctly, so its training accuracy is $654/918 = \mathbf{71.24\%}$. This number describes agreement on the same patients used to fit the rules. It does not estimate performance on new patients.

4.3 Descriptive analytics

Descriptive analytics finds structure without a target. *Clustering* groups observations with similar measurements. *Anomaly detection* marks observations that lie far from the rest.

Here, k-means clustering forms three groups of patients from `Age` and `MaxHR`, both rescaled to a common scale, and does not use `HeartDisease`. It then marks the five patients farthest from their group's centre. In decreasing order of distance they are CSV rows 830, 3, 508, 391, and 569, counted without the header. Figure 7 shows where they lie.

A flagged case is not a wrong case. Anomaly detection produces a ranked list for human attention. The method cannot say whether an unusual maximum heart rate is a data-entry error, a rare patient, or the most clinically interesting record in the table.

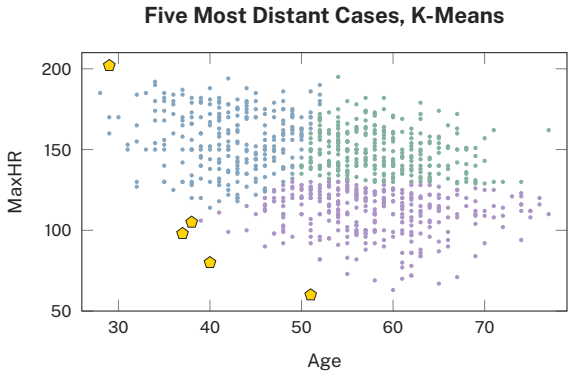


Figure 7. The three k-means groups are tinted blue, green, and purple, and the five observations selected for investigation are marked with gold pentagons. The axes retain the original measurement units.

5 CRISP-DM, the typical data science process

What it is for A project needs an order of work that connects the question to a useful result. CRISP-DM supplies that order, and it combines every idea from the previous sections.

The standard CRISP-DM is the **CR**oss **I**ndustry **S**tandard **P**rocess for **D**ata **M**ining [1]. In the Data Science Process Alliance's 2020 convenience poll of 109 respondents [7], CRISP-DM was the most selected option at 49% (Figure 8). The eight rounded percentages sum to 101%. A self-selected poll does not estimate prevalence among all practitioners.

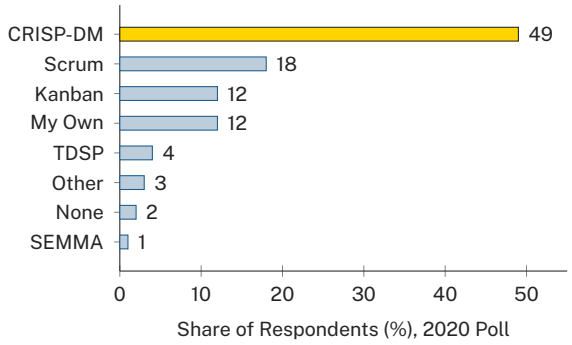


Figure 8. Processes selected in the Data Science Process Alliance's 2020 convenience poll, 109 respondents [7]

The six stages Figure 9 shows the stages. Each answers one question.

1. **Business understanding.** What are the business goals? How can data science help achieve them? Name the decision and who will use the result.
2. **Data understanding.** What data do we have or need? Are there data quality issues or limitations?
3. **Data preparation.** How do we clean and transform data for modelling? Document every change to units, missing values, and selected rows.

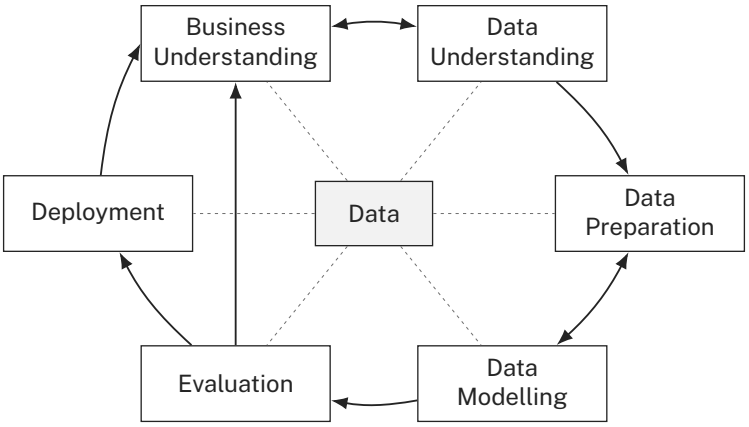


Figure 9. The six CRISP-DM stages. The cycle runs clockwise from business understanding, the two double-headed arrows mark the pairs that are worked back and forth, the arrow from evaluation to business understanding marks the return to the question, and every stage reads from and writes to the same data. After the CRISP-DM reference model of Chapman et al. [1].

4. **Data modelling.** What modelling techniques should we use?
5. **Evaluation.** Does the selected solution meet the business goals? This is a business question, not only an accuracy number. A model can score well and still fail the goal set in stage one.
6. **Deployment.** How do we deliver our solution to the business, and which deliverables give value to the people who use it?

Why it is a loop The double-headed arrows join business and data understanding, and data preparation and modelling. Each pair is worked out together. What the data can support limits which goal is realistic, so the question is revisited when the data cannot answer it. A method may need a different representation of the data, so modelling can send the work back to preparation. The arrow from evaluation to business understanding means that evaluation can return the project to its question rather than ending it or passing straight to deployment.

Worked example The Heart Disease examples fit onto the stages. Business understanding asks which patients need further investigation, and for whom. Data understanding is the import, the variable roles, and the exploratory plots. Data preparation includes decoding `FastingBS` and `HeartDisease` as categories. Data modelling is the fitted tree or the clustering. Evaluation asks whether 71.24% training accuracy is good enough for that clinical decision, and training accuracy alone cannot answer it. Deployment is a report or a tool that a clinician can use.

6 About the course

What it is for This section gives the course structure, tools, contacts, assessment, and dates.

6.1 The syllabus map

The tentative syllabus is drawn as one map whose spine is CRISP-DM. Business understanding, data understanding, data preparation, data modelling, evaluation, and deployment run left to right, with evaluation feeding back to business understanding. Four of those stages carry the course's technical content, listed in Table 3.

Table 3. Course topics mapped to CRISP-DM stages

Stage	Topics hanging from it
Data understanding	importing data; exploratory data analysis, split into summarizing and visualizing
Data preparation	cleaning; transforming; feature engineering; reducing dimensionality
Data modelling, predictive	tasks: classification, regression. Models: linear models, support vector machines, tree-based models. Assessment: performance metrics, performance estimation methods
Data modelling, descriptive	distance functions; clustering, split into partition-based, hierarchical, and density-based clustering; outlier detection
Deployment	reproducible reports; optional Shiny applications in R or Python

6.2 Learning outcomes for the course

By the end of the course, you should be able to

1. describe the main steps in a data science project;
2. transform data to prepare it for analysis and modelling;
3. explore data using appropriate statistics and visualizations;
4. design and implement a data science application, using predictive and descriptive modelling techniques;
5. evaluate, compare, and select models for a dataset; and
6. prepare reproducible reports in R Markdown or Python notebooks.

6.3 Technologies

Prepare one working R or Python environment. A free cloud notebook is sufficient, using Posit Cloud or Kaggle for R, or Posit Cloud, Kaggle, or Google Colab for Python. For local work, install R from CRAN with an editor such as RStudio, or Python 3 with a virtual environment and an editor such as VS Code or JupyterLab. Both languages use the same assessment requirements and rubric. Contact the instructor early if computer access is a barrier. Shiny applications in either language are an optional extension, not an additional assessed requirement.

This class uses base R, or Python with `pandas` and `matplotlib`. Free cloud services have usage caps. Save your files regularly and record the dependencies needed to reproduce the analysis.

6.4 Communication

- Announcements are posted in the course GitHub repository, `https://github.com/MAPS-Lab/CSCI-3141`. Course materials are public and can be read without signing in. You need a GitHub account for your assessment portfolio. Activate e-mail notifications for the repository and check your Dalhousie e-mail daily.
- Use Microsoft Teams for class questions and office-hour appointments.
- For private matters, use e-mail, with a subject line of the form `[CSCI 3141] Subject matter`.
- Instructor is Prof. Gabriel Spadon, `spadon@dal.ca`. Replies are normally within three business days.

Office hours are held **by appointment through Microsoft Teams**. Write ahead with the subject you would like to discuss. Individual and small-group appointments are available, so ask for a group appointment when several of you share a question. Bring a short, focused question, and ask early enough to use the feedback before your deadline.

6.5 Classes

Classes run Tuesdays and Thursdays, **16:05–17:25**, in Carleton Tupper Building room **3H01**, in person, across the twelve teaching weeks of the term. Class time includes ungraded questions, discussion, short worked examples, and work in pairs or small groups. There is no separate lab section; the integrated lab work begins in the Thursday classes.

6.6 Academic integrity and conduct

Academic integrity is expected from all students at Dalhousie. Review the Academic Integrity resources linked from the course GitHub repository. Originality checking follows syllabus A.9.1. Everyone at Dalhousie should be treated with dignity and respect, in class and online.

Generative AI

Generative AI is **prohibited on integrated labs** and examinations, and on assignments unless its use is expressly authorized in the assignment instructions. When it is authorized, follow the stated disclosure requirements.

6.7 Assessment

Table 4 lists the course assessment components, their weights, and the rule applied to each, and Table 5 describes the four assignments. Table 6 lists the key course dates and deadlines for fall 2026.

Table 4. Course assessment components and their weights

Component	Weight	Rule
Integrated labs	**15%**	best 7 of 8 results
Assignment 1	**7.5%**	individual
Assignment 2	**7.5%**	individual
Assignment 3	**15%**	individual
Assignment 4	**15%**	individual
Midterm	**15%**	in class, 20 October 2026
Final examination	**25%**	cumulative, Registrar-scheduled
Completion requirement	**Ungraded**	must be met to pass, syllabus A.7.4
Total	**100%**	

The weighted total, the combined assignment component, and the final examination must each reach 50%, and the assessment completion requirement in syllabus A.7.4 must be met, whatever the weighted percentage. The syllabus gives the full passing requirements and the absence provisions.

Integrated labs are marked with the common course rubric. Submit each lab before 16:05 Halifax time seven calendar days after it begins. Each lab handout states its date, and the deadline holds even when no class meets that day. Assignments are individual work, due at 23:59 on the announced date. Assignments 1 to 3 lose **5 percentage points per late calendar day**, up to seven days, after which later work requires an approved absence or accommodation process. Assignment 4 has no

Table 5. The four assignments

Assignment	Weight	What It Asks For	Due
Assignment 1	**7.5%**	Import data and perform basic data exploration	20 October 2026
Assignment 2	**7.5%**	Compare predictive models honestly	16 November 2026
Assignment 3	**15%**	Design and implement a predictive modelling solution	29 November 2026
Assignment 4	**15%**	Design and implement a descriptive modelling solution	9 December 2026

Table 6. Key course dates and deadlines for fall 2026

Date, 2026	Event	Kind
8 September	Classes begin	course
22 September	Last day to add	University
7 October	Last day to drop without a W	University
20 October	Midterm, in class	course
20 October	Assignment 1 due	course
16 November	Assignment 2 due	course
5 November	Last day to drop with a W	University
9–13 November	Fall Break	University
29 November	Assignment 3 due	course
3 December	Last class	course
9 December	Assignment 4 due	course
Registrar	Final examination	University

routine late window. Use the syllabus's equivalent private repository when accommodation, safety, privacy, copyright, or licensing requires it.

The midterm takes place in class on October 20, from 16:05 to 17:25, and the Registrar schedules the cumulative final examination. Bring valid Dalhousie photo ID. Course and personal notes are allowed unless the examination lists narrower approved aids. Internet access, communication, and generative AI are prohibited.

Build the portfolio in a private work repository with instructor access. Commit regularly, then record the full commit SHA and its commit URL in that item's `README.md` and push it by the deadline; the marker reads that README and assesses the snapshot it names. A branch link or movable tag alone does not fix the submitted version. Keep cloud notebooks and answer-revealing links private too. After the deadline and when the instructor opens sharing, publish that same commit and its history in your public portfolio and share that version at the next class meeting, if one follows.

Preserve assessment history permanently. Do not amend, rebase, squash, delete, replace, force-push, or backdate recorded commits, and do not move submitted tags. Keep submitted commits reachable. Record corrections as new commits, retaining the original submission. Git author dates alone are not deadline evidence; the pushed README record and instructor-retained snapshot provide the independent reference. Contact the instructor if an accidental disclosure requires correction.

Course materials released by the instructor may be shared with credit to the author and CSCI 3141 unless explicitly marked private or restricted. Preserve source links, third-party attributions, and licence terms. This permission does not authorize sharing another student's work or releasing assessed work early.

For an eligible short-term absence, contact the instructor before the requirement and submit the declaration within three calendar days after the last day of the absence, at most twice per course. Follow syllabus A.7.3 for other absences.

6.8 Course text and resources

There is **no required paid textbook**. The lecture slides, notes, and integrated lab handouts in the course GitHub repository are the primary material. Cheatsheets and documentation links are posted with each module.

7 Summary

- Data science combines computing, statistics, and domain knowledge to learn from observational and experimental data.
- This course uses tabular data, with n cases as rows and p variables as columns. The Heart Disease example has $n = 918$ cases and $p = 12$ attributes, and one row is one patient.
- Exploratory analysis summarizes and plots the sample without fitting a model.
- Predictive analytics learns a target, as in the tree splitting `MaxHR` at 133 and `Age` at 57. Its 71.24% training accuracy does not establish performance on new patients.
- Descriptive analytics finds structure without a target, as in the k-means example that marks five patients for investigation.
- CRISP-DM runs through business understanding, data understanding, data preparation, data modelling, evaluation, and deployment. Later evidence can revise earlier decisions.
- The course follows these stages, and its four assignments move from import to prepare to predict to describe. Portfolio work stays private until its deadline and the instructor's release.

References

[1] P. Chapman, J. Clinton, R. Kerber, T. Khabaza, T. Reinartz, C. Shearer, and R. Wirth. *CRISP-DM 1.0: Step-by-step Data Mining Guide*. Tech. rep. The CRISP-DM Consortium, 2000.

[2] D. Conway. *The Data Science Venn Diagram*. Reposted on 26 March 2013. Sept. 30, 2010. URL: `http://drewconway.com/zia/2013/3/26/the-data-science-venn-diagram`.

[3] D. Donoho. "50 years of data science". In: *Journal of Computational and Graphical Statistics* 26.4 (2017), pp. 745–766.

[4] fedesoriano. *Heart Failure Prediction Dataset*. Kaggle. Dataset, version 1. 2021.

[5] D. J. Hand, H. Mannila, and P. Smyth. *Principles of Data Mining*. MIT Press, 2001.

[6] A. Janosi, W. Steinbrunn, M. Pfisterer, and R. Detrano. *Heart Disease*. UCI Machine Learning Repository. Dataset. 1989.

[7] J. Saltz. *CRISP-DM is Still the Most Popular Framework for Executing Data Science Projects*. Data Science Process Alliance. Nov. 30, 2020. URL: `https://www.datascience-pm.com/crisp-dm-still-most-popular/`.